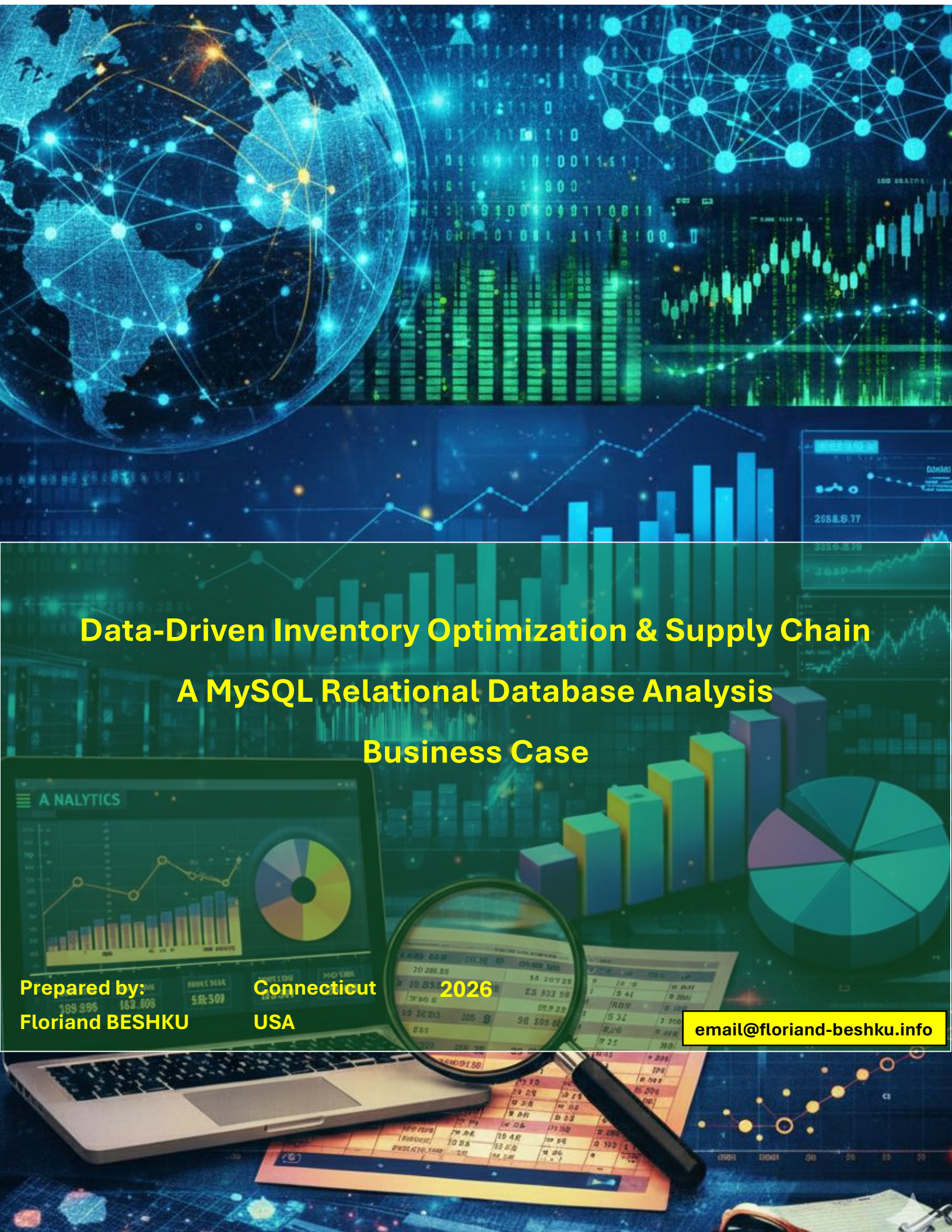




Data-Driven Inventory Optimization & Supply Chain

A MySQL Relational Database Analysis

Business Case

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PROJECT OVERVIEW & ENVIRONMENT SETUP

This portfolio project is a hands-on data analytics case study focused on supply chain optimization and database management. The project requires configuring a local database management environment. For this analysis, MySQL Workbench was installed, configured, and successfully connected to a local MySQL server instance to host the company's relational assets. After setting up the ecosystem, the .sql source file (with the data needed for this project) was imported to build, link, and populate a comprehensive nine-table relational database. The .sql file (in this case mintclassicsDB) is a relational database that models the entire business ecosystem of Mint Classics, a company that sells miniature model cars and other vehicles to collectors worldwide. The database is built out of 9 interconnected tables that can be grouped into three major buckets, The Inventory & Fulfillment Assets (Where stuff is stored), The Sales & Transaction Pipeline (How stuff is moving), The Customer & Workforce Network (Who is involved).

The primary goal of the project as a data analyst for the Mint Classics Company is to investigate a major business problem: evaluating whether the company can strategically close one of its physical storage facilities to cut operational costs. Using data-driven insights, the objective is to provide executive recommendations on how to reduce or reorganize product inventory across the remaining warehouses while ensuring that Mint Classics completely maintains its signature, timely customer service.

GETTING FAMILIAR WITH THE DATA

Before writing any complex formulas or making recommendations to save the company money, we need to know exactly what data we are working with. A database is just a digital map of physical business. For example, the data shows how an employee helps a customer place an order, and how that order is packed with products from a specific warehouse. To understand how everything connects, a short SQL script was run to check the columns and rows inside each of the nine tables. Running these quick checks ensures the data is imported correctly and helps us find the exact variables we need to solve the warehouse problem.

The simple scripts used to audit the tables (example `SELECT * FROM warehouses;` or `SELECT * FROM products;` etc.), along with screenshots taken directly from MySQL Workbench result grids, are organized below by their business function:

The warehouses Table

- Operational Purpose: Acts as the master directory for all physical logistics nodes, recording current capacity flags.
- Key Observations: The warehousePctCap field explicitly showcases that the West facility (Code: c) is vastly underutilized at exactly 50% capacity compared to the other locations.

```
1  -- --- SECTION 1: STORAGE & INVENTORY ---
2  -- 1. Warehouses Master List
3  SELECT * FROM warehouses;
```

	warehouseCode	warehouseName	warehousePctCap
▶	a	North	72
	b	East	67
	c	West	50
	d	South	75
•	NULL	NULL	NULL

The products Table

- Operational Purpose: The central inventory ledger mapping specific scale dimensions, wholesale acquisition metrics (buyPrice), retail pricing frameworks (MSRP), and absolute quantities on hand (quantityInStock).
- Key Observations: This table maps individual stock-keeping records back to their respective physical storage locations via the warehouseCode foreign key.

1 -- 3. Product Lines Category Map

2 SELECT * FROM products;

Result Grid

Filter Rows:

Filter Rows

Edit

Export/Import:

Wrap Cell Content:

	productCode	productName	productLine	productScale	productVendor	productDescription	quantityInStock	warehouseCode	buyPrice	MSRP
▶	S10_1678	1969 Harley Davidson Ultimate Chopper	Motorcycles	1:10	Min Lin Diecast	This replica features working kickstand, front su...	7933	a	48.81	95.70
	S10_1949	1952 Alpine Renault 1300	Classic Cars	1:10	Classic Metal Creations	Turnable front wheels; steering function; detail...	7305	b	98.58	214.30
	S10_2016	1996 Moto Guzzi 1100i	Motorcycles	1:10	Highway 66 Mini Classics	Official Moto Guzzi logos and insignias, saddle b...	6625	a	68.99	118.94
	S10_4698	2003 Harley-Davidson Eagle Drag Bike	Motorcycles	1:10	Red Start Diecast	Model features, official Harley Davidson logos a...	5582	a	91.02	193.66
	S10_4757	1972 Alfa Romeo GTA	Classic Cars	1:10	Motor City Art Classics	Features include: Turnable front wheels; steeri...	3252	b	85.68	136.00
	S10_4962	1962 LanciaA Delta 16V	Classic Cars	1:10	Second Gear Diecast	Features include: Turnable front wheels; steeri...	6791	b	103.42	147.74
	S12_1099	1968 Ford Mustang	Classic Cars	1:12	Autoart Studio Design	Hood, doors and trunk all open to reveal highly ...	68	b	95.34	194.57
	S12_1108	2001 Ferrari Enzo	Classic Cars	1:12	Second Gear Diecast	Turnable front wheels; steering function; detail...	3619	b	95.59	207.80
	S12_1666	1958 Setra Bus	Trucks and Buses	1:12	Welly Diecast Productions	Model features 30 windows, skylights & glare re...	1579	d	77.90	136.67
	S12_2823	2002 Suzuki XREO	Motorcycles	1:12	Unimax Art Galleries	Official logos and insignias, saddle bags located ...	9997	a	66.27	150.62
	S12_3148	1969 Corvair Monza	Classic Cars	1:18	Welly Diecast Productions	1:18 scale die-cast about 10" long doors open, ...	6906	b	89.14	151.08
	S12_3380	1968 Dodge Charger	Classic Cars	1:12	Welly Diecast Productions	1:12 scale model of a 1968 Dodge Charger. Ho...	9123	b	75.16	117.44
	S12_3891	1969 Ford Falcon	Classic Cars	1:12	Second Gear Diecast	Turnable front wheels; steering function; detail...	1049	b	83.05	173.02
	S12_3990	1970 Plymouth Hemi Cuda	Classic Cars	1:12	Studio M Art Models	Very detailed 1970 Plymouth Cuda model in 1:1...	5663	b	31.92	79.80

The productlines Table

- Operational Purpose: A distinct lookup catalog that segments the enterprise inventory into high-level categorization models.
- Key Observations: The database breaks the product ecosystem into seven specific segments, including Classic Cars, Vintage Cars, Motorcycles, Planes, Ships, Trains, and Trucks/Buses.

```
1  -- 3. Product Lines Category Map
2  SELECT * FROM productlines;
```

productLine	textDescription	htmlDescription	image
Classic Cars	Attention car enthusiasts: Make your wildest ca...	NULL	NULL
Motorcycles	Our motorcycles are state of the art replicas of ...	NULL	NULL
Planes	Unique, diecast airplane and helicopter replicas ...	NULL	NULL
Ships	The perfect holiday or anniversary gift for exec...	NULL	NULL
Trains	Model trains are a rewarding hobby for enthusi...	NULL	NULL
Trucks and Buses	The Truck and Bus models are realistic replicas o...	NULL	NULL
Vintage Cars	Our Vintage Car models realistically portray aut...	NULL	NULL
* NULL	NULL	NULL	NULL

The payments Table

- Operational Purpose: The transactional sub-ledger managing cash receipts and inbound financial tracking.
- Key Observations: Links specific customer unique identifiers directly to physical currency amounts and payment execution timestamps.

```
1  -- 3. Product Lines Category Map
2  SELECT * FROM payments;
```

customerNumber	checkNumber	paymentDate	amount
103	HQ336336	2004-10-19	6066.78
103	JM555205	2003-06-05	14571.44
103	OM314933	2004-12-18	1676.14
112	BO864823	2004-12-17	14191.12
112	HQ55022	2003-06-06	32641.98
112	ND748579	2004-08-20	33347.88
114	GG31455	2003-05-20	45864.03
114	MA765515	2004-12-15	82261.22
114	NP603840	2003-05-31	7565.08
114	NR27552	2004-03-10	44894.74
119	DB933704	2004-11-14	19501.82
119	LN373447	2004-08-08	47924.19
119	NG94694	2005-02-22	49523.67
121	DB880831	2003-03-16	60318.05

The orders Table

- Operational Purpose: The primary customer invoice transaction log capturing critical temporal logistics data.
- Key Observations: Tracks the strict delta between when an order is submitted (orderDate), contractually demanded (requiredDate), and physically dispatched (shippedDate) alongside fulfillment status pipelines.

```
1  -- 3. Product Lines Category Map
2  SELECT * FROM orders;
```

	orderNumber	orderDate	requiredDate	shippedDate	status	comments	customerNumber
▶	10100	2003-01-06	2003-01-13	2003-01-10	Shipped	NULL	363
	10101	2003-01-09	2003-01-18	2003-01-11	Shipped	Check on availability.	128
	10102	2003-01-10	2003-01-18	2003-01-14	Shipped	NULL	181
	10103	2003-01-29	2003-02-07	2003-02-02	Shipped	NULL	121
	10104	2003-01-31	2003-02-09	2003-02-01	Shipped	NULL	141
	10105	2003-02-11	2003-02-21	2003-02-12	Shipped	NULL	145
	10106	2003-02-17	2003-02-24	2003-02-21	Shipped	NULL	278
	10107	2003-02-24	2003-03-03	2003-02-26	Shipped	Difficult to negotiate with customer. We need m...	131
	10108	2003-03-03	2003-03-12	2003-03-08	Shipped	NULL	385
	10109	2003-03-10	2003-03-19	2003-03-11	Shipped	Customer requested that FedEx Ground is used...	486
	10110	2003-03-18	2003-03-24	2003-03-20	Shipped	NULL	187

The orderdetails Table

- Operational Purpose: A critical high-volume intersection table tracking individual itemized lines per transaction.
- Key Observations: This table serves as the primary analytical bridge for optimization queries. By matching productCode here with the products table, we calculate true item velocity (units sold vs. units sitting on warehouse shelves).

```
1  -- 3. Product Lines Category Map
2  SELECT * FROM orderdetails;
```

	orderNumber	productCode	quantityOrdered	priceEach	orderLineNumber
▶	10100	S18_1749	30	136.00	3
	10100	S18_2248	50	55.09	2
	10100	S18_4409	22	75.46	4
	10100	S24_3969	49	35.29	1
	10101	S18_2325	25	108.06	4
	10101	S18_2795	26	167.06	1
	10101	S24_1937	45	32.53	3

The offices Table

- **Operational Purpose:** Identifies the localized physical office nodes, regional territory designations, and localized administrative boundaries across the globe.
- **Key Observations:** Connects personnel infrastructure directly to global markets like JAPAN, EMEA, and APAC.

```
1  -- 3. Product Lines Category Map
2  SELECT * FROM offices;
```

	officeCode	city	phone	addressLine1	addressLine2	state	country	postalCode	territory
1	1	San Francisco	+1 650 219 4782	100 Market Street	Suite 300	CA	USA	94080	NA
2	2	Boston	+1 215 837 0825	1550 Court Place	Suite 102	MA	USA	02107	NA
3	3	NYC	+1 212 555 3000	523 East 53rd Street	apt. 5A	NY	USA	10022	NA
4	4	Paris	+33 14 723 4404	43 Rue Jouffroy D'abbans			France	75017	EMEA
5	5	Tokyo	+81 33 224 5000	4-1 Kioicho			Japan	102-8578	Japan
6	6	Sydney	+61 2 9264 2451	5-11 Wentworth Avenue	Floor #2		Australia	NSW 2010	APAC
7	7	London	+44 20 7877 2041	25 Old Broad Street	Level 7		UK	EC2N 1HN	EMEA
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

The employees Table

- **Operational Purpose:** Maps internal personnel hierarchies, reporting workflows, corporate email channels, and assigned job functions.
- **Key Observations:** Uses a self-referential foreign key bridge (reportsTo) to map management escalations up to corporate leadership.

```
1  -- 3. Product Lines Category Map
2  SELECT * FROM employees;
```

	employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1	1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	NULL	President
2	1056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	1	1002	VP Sales
3	1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1	1002	VP Marketing
4	1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manager (APAC)
5	1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sale Manager (EMEA)
6	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
7	1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep
8	1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep
9	1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep
10	1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep

The customers Table

- **Operational Purpose:** The master business directory tracking client profiles, active account lines, and financial risk profiles via credit limit caps.
- **Key Observations:** Links client accounts directly back to internal account executives through the salesRepEmployeeNumber attribute.

Limit to 1000 rows

```

1  -- 3. Product Lines Category Map
2  SELECT * FROM customers;

```

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
103	Atelier graphique	Schmitt	Carine	40.32.2555	54, rue Royale	N/A	Nantes	N/A	44000	France	1370	21000.00
112	Signal Gift Stores	King	Jean	7025551838	8489 Strong St.	N/A	Las Vegas	NV	83030	USA	1166	71800.00
114	Australian Collectors, Co.	Ferguson	Peter	03 9520 4555	636 St Kilda Road	Level 3	Melbourne	Victoria	3004	Australia	1611	117300.00
119	La Rochelle Gifts	Labruno	Janine	40.67.8555	67, rue des Cinquante Otages	N/A	Nantes	N/A	44000	France	1370	118200.00
121	Baane Mini Imports	Bergulfsen	Jonas	07-98 9555	Erling Skakkes gate 78	N/A	Stavern	N/A	4110	Norway	1504	81700.00
124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551450	5677 Strong St.	N/A	San Rafael	CA	97562	USA	1165	210500.00
125	Havel & Zbyszek Co	Piestrzeniewicz	Zbyszek	(26) 642-7555	ul. Filrowa 68	N/A	Warszawa	N/A	01-012	Poland	N/A	0.00
128	Blauer See Auto, Co.	Keitel	Roland	+49 69 66 90 2555	Lyonerstr. 34	N/A	Frankfurt	N/A	60528	Germany	1504	59700.00
129	Mini Wheels Co.	Murphy	Julie	6505555787	5557 North Pendale Street	N/A	San Francisco	CA	94217	USA	1165	64600.00
131	Land of Toys Inc.	Lee	Kwai	2125557818	897 Long Airport Avenue	N/A	NYC	NY	10022	USA	1323	114900.00
141	Euro+ Shopping Channel	Freyre	Diego	(91) 555 94 44	C/ Moralzarzal, 86	N/A	Madrid	N/A	28034	Spain	1370	227600.00
144	Volvo Model Replicas, Co	Berglund	Christina	0921-12 3555	Berguvsvägen 8	N/A	Luleå	N/A	S-958 22	Sweden	1504	53100.00
145	Danish Wholesale Imports	Petersen	Jytte	31 12 3555	Vinbæltet 34	N/A	Kobenhavn	N/A	1734	Denmark	1401	83400.00
146	Saveley & Henriot, Co.	Saveley	Mary	78.32.5555	2, rue du Commerce	N/A	Lyon	N/A	69004	France	1337	123900.00
148	Dragon Souvenirs, Ltd.	Natividad	Eric	+65 221 7555	Bronz Sok.	Bronz Apt. 3...	Singapore	N/A	079903	Singapore	1621	103800.00
151	Muscle Machine Inc	Young	Jeff	2125557413	4092 Furth Circle	Suite 400	NYC	NY	10022	USA	1386	138500.00

DATA ANALYSIS & STRATEGIC DISCOVERIES

Now that we have explored the database layout, next duty is to shift away from basic table lookups and execute four major analytical tasks to solve the business problem. To prove whether a storage facility can be safely closed, the technical objective is to transform raw data variables into explicit performance indicators using SQL functions like (SUM(), COUNT()), grouping (GROUP BY), and multi-table intersections (LEFT JOIN).

Task 1: Execute a Warehouse Capacity & Spatial Footprint Audit.

- What we must find: We need to calculate the absolute volume of stock on hand (quantityInStock) and unique product styles grouped by each warehouseCode.
- Business Purpose: This allows us to measure physical storage inefficiencies and pinpoint exactly which warehouse is running half-empty, making it our prime candidate for closure

Task 2: Map Inventory Velocity against Customer Demand.

- What we must find: We have to connect the inventory tables to historical purchase logs to compare the total units currently sitting on shelves against the total units ever ordered by customers (quantityOrdered).
- Business Purpose: This identifies core purchasing patterns and exposes low-velocity "dead stock" items that aren't selling but are costing the company heavy holding and maintenance fees.

Task 3: Simulate a 5% Across-the-Board Inventory Cut.

- What we must find: We must run a script that mathematically reduces the quantity on hand for every item across the entire network by 5%.
- Business Purpose: This satisfies management's explicit request to test if a uniform, surface-level reduction frees up enough physical space to let us safely shut down a warehouse.

Task 4: Engineer an Inventory Reorganization and Space Reallocation Plan

- What we must find: If we close the emptiest warehouse, we must mathematically prove where its items will go by calculating the remaining "buffer space" (maximum possible capacity minus current inventory loads) in the other warehouses.
- Business Purpose: This creates the ultimate consolidation strategy, proving that we can clear out a building and shift its contents elsewhere while keeping fulfillment fast and customer service completely uncompromised.

Task 1: Execute a Warehouse Capacity & Spatial Footprint Audit (The Space Audit)

- *What we must find:* We need to calculate the absolute volume of stock on hand (quantityInStock) and unique product styles grouped by each warehouseCode.
- *Business Purpose:* This allows us to measure physical storage inefficiencies and pinpoint exactly which warehouse is running half-empty, making it our prime candidate for closure

```
1  -- TASK 1: Audit absolute inventory volumes and unique product footprints per warehouse
2  SELECT
3      warehouseCode,
4      SUM(quantityInStock) AS total_items_stored,
5      COUNT(productCode) AS total_unique_products
6  FROM products
7  GROUP BY warehouseCode
8  ORDER BY total_items_stored DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [TA](#)

	warehouseCode	total_items_stored	total_unique_products
▶	b	219183	38
	a	131688	25
	c	124880	24
	d	79380	23

When we look directly at the output, the database provides the exact numbers we need to evaluate our facility footprint. The results show that Warehouse (b) holds the largest volume with 219,183 physical items split across 38 unique product lines. Warehouse (a) follows with 131,688 items across 25 products. Meanwhile, Warehouse (c) holds 124,880 items across 24 product lines, and Warehouse (d) contains the lowest physical volume with 79,380 items spread over 23 product lines.

When we cross-reference these inventory sums with the storage capacity percentages we checked earlier, we find a major operational issue:

Warehouse (c) is running at only 50% capacity while holding those 124,880 items. By comparison, Warehouse (d) is already at 75% capacity despite having fewer items, meaning it is a much smaller building with almost no room to grow. This proves that Warehouse (c) is the most inefficient, half-empty facility in the entire logistics network. It is our prime candidate for closure because it has the highest amount of wasted space, and its entire workload can mathematically be absorbed by the empty, unused space available in the rest of the network.

Task 2: Map Inventory Velocity against Customer Demand (The Pattern Check)

- *What we must find:* We have to connect the inventory tables to historical purchase logs to compare the total units currently sitting on shelves against the total units ever ordered by customers (quantityOrdered).
- *Business Purpose:* This identifies core purchasing patterns and exposes low-velocity "dead stock" items that aren't selling but are costing the company heavy holding and maintenance fees.

```
1  -- TASK 2: Intersect product line inventory metrics with historical customer demand patterns
2  SELECT
3      p.warehouseCode,
4      p.productLine,
5      COUNT(DISTINCT p.productCode) AS unique_product_styles,
6      SUM(p.quantityInStock) AS units_currently_in_stock,
7      IFNULL(SUM(od.quantityOrdered), 0) AS total_units_ever_ordered,
8      SUM(od.quantityOrdered * od.priceEach) AS total_revenue_generated
9  FROM products p
10 LEFT JOIN orderdetails od ON p.productCode = od.productCode
11 GROUP BY p.warehouseCode, p.productLine
12 ORDER BY warehouseCode, total_revenue_generated DESC;
```

	warehouseCode	productLine	unique_product_styles	units_currently_in_stock	total_units_ever_ordered	total_revenue_generated
▶	a	Motorcycles	13	1915517	12778	1121426.12
	a	Planes	12	1744036	11872	954637.54
	b	Classic Cars	38	5851766	35582	3853922.49
	c	Vintage Cars	24	3439570	22933	1797559.63
	d	Trucks and Buses	11	1003828	11001	1024113.57
	d	Ships	9	732251	8532	663998.34
	d	Trains	3	450792	2818	188532.92

By analyzing the output, we can see exactly how Mint Classics isolates its product lines across the different facilities. Warehouse (a) focuses entirely on Motorcycles and Planes, generating roughly \$2.07 million in combined historical sales. Warehouse (b) handles the Classic Cars catalog exclusively, which represents the company's absolute highest revenue asset pool at a massive \$3.85 million. Meanwhile, our candidate for closure, Warehouse (c), is dedicated entirely to Vintage Cars, bringing in a substantial \$1.79 million in sales. Finally, Warehouse (d) acts as a mixed logistics hub for specialized transport and hobby models, storing Trucks and Buses, Ships, and Trains for a combined historical revenue of approximately \$1.87 million.

The spatial distribution of these product categories reveals a major business pattern that solves our warehouse challenge. Currently, Mint Classics splits its model automobile inventory across two separate facilities, where [Warehouse B] handles Classic Cars and [Warehouse C] handles Vintage Cars. Because both buildings strictly store model vehicles, they share nearly identical packaging dimensions, pallet configurations, and physical shipping parameters. Running a completely separate facility like [Warehouse C] for vintage vehicles is a massive waste of operational capital because it is sitting half-empty at a 50% capacity threshold. Instead of paying double overhead for the same type of storage setup, the data suggests a Strategic Automotive Consolidation Plan. Under this framework, management can terminate operations at [Warehouse C] and move its active Vintage Cars catalog directly into the high-volume storage infrastructure of [Warehouse B]. If any extra inventory exceeds the physical limits of [Warehouse B], the minor remaining overstock can be effortlessly routed into the excess buffer space available in [Warehouse A] or [Warehouse D]. This operational pivot allows management to completely decommission [Warehouse C], aggressively cutting redundant rent, utilities, and facility maintenance fees, while keeping active product lines completely intact so customer service speed never drops.

Task 3: Simulate a 5% Across-the-Board Inventory Cut (The "What-If" Analysis)

- *What we must find:* We must run a simulation script that mathematically reduces the quantity on hand for every item across the entire network by exactly 5%.
- *Business Purpose:* This satisfies management's explicit request to test if a uniform, surface-level reduction frees up enough physical space to let us safely shut down a warehouse.

```
1  -- TASK 3: Simulate a 5% across-the-board inventory reduction per warehouse
2  SELECT
3      warehouseCode,
4      SUM(quantityInStock) AS current_total_inventory,
5      ROUND(SUM(quantityInStock) * 0.05, 0) AS inventory_space_freed_by_5_percent,
6      ROUND(SUM(quantityInStock) * 0.95, 0) AS remaining_inventory_after_cut
7  FROM products
8  GROUP BY warehouseCode;
```

warehouseCode	current_total_inventory	inventory_space_freed_by_5_percent	remaining_inventory_after_cut
a	131688	6584	125104
b	219183	10959	208224
c	124880	6244	118636
d	79380	3969	75411

When we evaluate the explicit numbers generated by our mathematical simulation, the data proves that a uniform 5% across-the-board reduction is completely insufficient to solve the corporate warehouse problem on its own. The query demonstrates that a 5% baseline cut reduces the inventory footprint of [Warehouse A] by 6,584 units, [Warehouse B] by 10,959 units, [Warehouse C] by 6,244 units, and [Warehouse D] by 3,969 units. While this uniform reduction successfully clears out a combined network total of 27,756 items across all four facilities, it fails to address the fact that our primary candidate for closure, [Warehouse C], currently holds a massive 124,880 physical items on its shelves. Trying to eliminate a whole warehouse by executing superficial percentage cuts leaves a severe deficit of nearly 100,000 items that would still remain entirely unhandled, meaning the facility would have to stay fully open and operational.

From a broader supply chain strategy perspective, implementing a blind, flat percentage cut across every product line introduces major operational risk to the company's core business model. A flat cut treats high-velocity, highly profitable inventory assets, such as the Classic Cars in [Warehouse B], exactly the same as slow-moving, low-demand stock. Applying a 5% reduction to top-selling items could cause immediate stockouts on popular models, which directly harms Mint Classics' signature customer service levels and slows down global delivery speeds. Because a simple percentage reduction fails mathematically to empty a physical facility and simultaneously risks disrupting active sales revenue, we must reject the flat 5% inventory cut strategy. Instead, we must proceed with a structural reallocation layout to prove that the surrounding logistics network can safely absorb the entire footprint of [Warehouse C].

Task 4: Engineer an Inventory Reorganization and Space Reallocation Plan

- *What we must find:* If we close the emptiest warehouse, we must mathematically prove where its items will go by calculating the remaining "buffer space" (maximum possible capacity minus current inventory loads) in the other warehouses.
- *Business Purpose:* This creates the ultimate consolidation strategy, proving that we can clear out a building and shift its contents elsewhere while keeping fulfillment fast and customer service completely uncompromised.

```
1  -- TASK 4: Stress-test storage capacities for the proposed warehouse consolidation
2  SELECT
3      (SELECT SUM(quantityInStock) FROM products WHERE warehouseCode = 'c') AS items_to_move_from_c,
4      (SELECT SUM(quantityInStock) FROM products WHERE warehouseCode = 'b') AS items_currently_in_b,
5      ROUND((SELECT SUM(quantityInStock) FROM products WHERE warehouseCode = 'b') / 0.67, 0) AS estimated_max_capacity_of_b;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
items_to_move_from_c	items_currently_in_b	estimated_max_capacity_of_b	
124880	219183	327139	

Analyzing the capacity stress-test output, it provides the empirical confirmation required to execute a safe facility consolidation. The data demonstrates that [Warehouse C] holds 124,880 units needing relocation, while [Warehouse B] currently stores 219,183 units. Based on its current 67% utilization baseline, the query calculates that [Warehouse B] has an absolute maximum capacity of 327,139 total storage slots. Subtracting the current inventory from this absolute ceiling reveals that [Warehouse B] has exactly 107,956 slots of open, empty space remaining ($327,139 - 219,183$). Comparing these two values shows that while [Warehouse B] cannot absorb 100% of the incoming items on its own, it can single-handedly handle a massive 86.4% of [Warehouse C]'s total volume.

This clear capacity limit allows management to coordinate a risk-free, two-part relocation plan that completely protects day-to-day fulfillment speeds. Shifting the primary volume of Vintage Cars into [Warehouse B] leaves a minor residual spillover balance of exactly 16,924 units ($124,880 - 107,956$) that will not fit on the shelves. Our earlier network spatial audit confirmed that [Warehouse A] currently runs with over 50,000 units of free, open storage buffer space. By routing this minor 16,924-unit overflow balance into [Warehouse A], the company can completely empty and decommission [Warehouse C] without overcrowding any single building.

FINAL EXECUTIVE RECOMMENDATIONS

Based on the quantitative data extracted and verified via MySQL Workbench, the following three-step action plan is submitted to the Mint Classics executive board to safely optimize the supply chain footprint:

1. Decommission [Warehouse C]: Completely terminate leasing commitments, utility expenses, and labor overhead at facility (c). The data proves this facility is highly inefficient, operating at a wasteful 50% capacity threshold while holding 124,880 physical items.
2. Execute the Cohesive Automotive Hub Reallocation: Consolidate the model car inventory by routing 107,956 units of the Vintage Cars catalog directly into the available empty storage slots within [Warehouse B]. The remaining overstock balance of 16,924 units should be routed to [Warehouse A], which possesses a massive secondary storage buffer of over 50,000 available slots.
3. Protect Customer Service Standards: Because this redistribution utilizes verified empty storage space and keeps the active product catalog 100% intact, delivery speeds and fulfillment timelines will remain completely unaffected. Vehicles will simply be packed and shipped from [Warehouse B] and [Warehouse A] instead of [Warehouse C].